A Multitask Framework for Hyperspectral Change Detection and Band Reweighting With Unbalanced Contrastive Learning

Xiande Wu, Paolo Gamba, *Fellow, IEEE*, Jie Feng, *Senior Member, IEEE*, Ronghua Shang, *Senior Member, IEEE*, Xiangrong Zhang, *Senior Member, IEEE*, and Licheng Jiao, *Fellow, IEEE*

***Abstract*— Multitask learning has been widely applied in visual learning to significantly enhance the performance. The combination of hyperspectral change detection (HCD) and band reweighting can achieve discriminative feature enhancement for improving detection performance. However, existing multitask models for these two tasks are unidirectional, with band reweighting unable to learn from task guidance. To address this challenge, a multitask HCD (MHCD) framework with differential band reweighting and unbalanced contrastive learning is proposed. MHCD consists of a differential band reweighting network (DBRN) and a Siamese detection network. DBRN extracts discriminative information for HCD by analyzing the differential spatial-spectral information across time states, whose optimization is under the guidance of HCD. Furthermore, a multitemporal interaction module and multidomain fusion module are inserted into the Siamese detection network. They hierarchically connect cross-temporal features and fuse features from spatial, spectral, and temporal domains, providing complementary clues in these different domains. Considering the sample imbalance and enormous variation within a class in binary HCD, an unbalanced contrastive learning method based on multiple prototypes (UCLM) tailored has been considered. It estimates multiple prototypes to flexibly adjust the contribution of different classes of samples to the loss. The proposed method has been validated using three public benchmark datasets, demonstrating improvements in multiple metrics for change detection. The code of our paper is available at: https://github.com/jiefeng0109/MHCD.**

***Index Terms*— Band reweighting, change detection, deep learning, hyperspectral image (HSI), multitask learning.**

Manuscript received 19 March 2024; revised 6 May 2024; accepted 2 June 2024. Date of publication 5 June 2024; date of current version 20 September 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62271374, Grant 62176200, Grant 62077038, and Grant 62176196; in part by the State Key Program of National Natural Science of China under Grant 61836009; in part by the Natural Science Basic Research Program of Shaanxi under Grant 2022GY-065 and Grant 2022JC-45; in part by the Fundamental Research Funds for the Central Universities under Grant QTZX23047; and in part by the China Scholarship Council. *(Corresponding author: Jie Feng.)*

Xiande Wu, Jie Feng, Ronghua Shang, Xiangrong Zhang, and Licheng Jiao are with the Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education of China, Xidian University, Xi'an 710071, China (e-mail: xdwu95@gmail.com; jiefeng0109@163.com; rhshang@mail.xidian.edu.cn; xrzhang@mail.xidian.edu.cn; lchjiao@mail.xidian.edu.cn).

Paolo Gamba is with the Telecommunications and Remote Sensing Laboratory of Italy, University of Pavia, 27100 Pavia, Italy (e-mail: paolo.gamba@unipv.it).

Digital Object Identifier 10.1109/TGRS.2024.3410131

I. Introduction

CHANGE detection plays a crucial role in the dynamic observation of the Earth, particularly when dealing with disaster response [1], [2] and land cover monitoring [3], [4], [5], [6]. Initially, change detection techniques were predominantly applied to synthetic aperture radar and multispectral images. However, with continuous innovation in remote sensing technology, the acquisition and processing capabilities of hyperspectral images (HSIs) have significantly improved, making them a mainstream research focus. In comparison to other remote sensing images, HSIs with hundreds of bands can provide more abundant and detailed spectral information, thereby enabling more accurate and comprehensive interpretation and detection of surface changes [7].

In the realm of HSI change detection, diverse methodologies have been explored, falling into three main categories: image algebra methods [8], [9], [10], image transformation methods [11], [12], [13], and image classification methods [14], [15]. While image algebra methods, such as change vector analysis (CVA) [8] and its compressed variant (C2VA) [9], offer simplicity and efficiency, they lack spatial information sensitivity. Image transformation methods, such as principal component analysis (PCA) [12] and minimum average difference (MAD) [13]. These methods transform the original image into a new space and achieve better detection by highlighting the changed information. However, most of them face challenges in selecting appropriate thresholds. Image classification methods, including post-classification and compound classification, provide multiple change information but are limited by the classifier performance, especially when dealing with complex changes. Finally, in most of these methods manual parameter selection is required, a task that may be cumbersome and time-consuming.

Lately, and mostly because of their automatic feature extraction ability, deep learning techniques have sparked significant interest in HSI change detection. Indeed, deep learning methods, such as deep belief networks (DBNs) [16], [17] and autoencoders (AEs) [18], [19], have been introduced to HSI processing, but originally they focused on spectral features. In order to enhance performances, other network architectures, such as convolutional neural networks (CNNs) [20], [21], [23], recurrent neural networks (RNNs) [24], [25], [26], and transformers [27], [28], [29], [30], [31] have been designed to extract highly discriminative spatial-spectral features for change detection. For instance, in [32] a dual-stream encoder–decoder was introduced. It integrates hierarchical

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:49:48 UTC from IEEE Xplore. Restrictions apply.

temporal features extracted by two encoders and fuses them for detection. Alternatively, attention mechanisms have been applied in many network structures to enhance focus on specific information. As an example, Wang et al. [33] presented a Siamese network based on spatial-spectral attention. Learning weights in both spatial and spectral domains emphasizes informative channels and locations. As a third alternative, due to their advantages in modeling long-term temporal information, RNNs are often employed for modeling temporal relationships between different phases in change detection. Song et al. [24] and Shi et al. [25] fused CNNs and long short-term memory (LSTM) models, utilizing LSTM to analyze temporal relationships among multitemporal HSIs. In [34], a time-distance-guided LSTM is proposed for change detection in irregularly collected images. Additionally, a focal weighted cross-entropy loss is introduced here, which improves the network's performance on hard examples and imbalanced data. Finally, transformer-based methods have been introduced for change detection. Transformers extract global context information from HSIs, achieving accurate change detection results. In [30], a spectral, spatial, and temporal transformer is proposed. Together, they work to effectively characterize the 3-D structure of HSIs. Considering the drawback of requiring a large number of samples, Wang et al. [31] proposed a semi-supervised transformer AE. It can align the domains of different images to distinguish changing pixels.

In addition to structural improvements, some studies focus on applying untrained models to change detection. In [35], a novel framework for HSI change detection is proposed, utilizing untrained models to extract bitemporal features. Then deep change hypervectors are employed to cluster the changed pixels into different groups. To improve the adaptability of untrained models, Yang et al. [36] introduced dilation-deformable feature extraction to extract multiscale spatial–spectral features over unfixed sampling locations.

Multitask learning enhances model performance through information sharing, a principle well-established in natural language processing, computer vision, and speech recognition. Specifically, the tasks of hyperspectral band reweighting and hyperspectral change detection (HCD) are closely related, as the accurate choice of bands can provide more sensitive and discriminative features, aiding in the precise detection of surface changes. Therefore, combining these two tasks in a multitask learning framework is feasible. As a special case of band reweighting, band selection and HCD are combined as a multitask learning framework. Liu et al. [37] treated band selection as a preprocessing step, analyzing the change detection performance after band selection. Here, two tasks are separated to be optimized. In [38], an HCD algorithm based on slow-fast band selection was proposed. It integrates CNN to extract more discriminative change and no-change features by selecting bands with different change rates. While these methods have provided initial validation of the correlation between change detection and band selection tasks, they are unidirectional frameworks where band selection serves as a preprocessing step and does not jointly learn with change detection to adapt to different tasks.

In this article, we propose a multitask HCD (MHCD) framework incorporating novel tasks of differential band reweighting and unbalanced contrastive learning. In MHCD, a differential band reweighting network (DBRN) extracts differential spatial-spectral information to select the most sensitive bands for HCD. To enhance temporal interactions across multiple time steps, we also introduce a multitemporal interaction module (MTIM) and a multidomain fusion module (MDFM). The MTIM is embedded in a Siamese detection network to interactively capture features across multiple temporal phases, while MDFM is employed at the final stage to fuse features from different domains, leveraging their diverse discriminative power. Together, these novel modules provide complementary information in terms of time, space, and spectral aspects. Furthermore, we introduce an unbalanced contrastive learning method based on multiple prototypes (UCLM). The use of multiple prototypes allows for a more flexible selection of the sample contrasts, adapting to variations among samples of the same class. Concurrently, an enhanced contrastive loss is employed to adjust the contribution of different classes, mitigating the training instability induced by imbalanced training data. Accordingly, the end-to-end optimization of the MHCD is performed by jointly considering the contrastive loss from UCLM and the classification loss.

The main contributions of this article can thus be summarized as follows:

1) In MHCD, change detection and band reweighting are learned via an innovative multitask learning approach. Change detection facilitates band reweighting with discriminative ability, while band reweighting enhances change detection by utilizing temporal differencing information.
2) The integration of the two novel proposed modules (MTIM and MDFM) into the detection network not only facilitates hierarchical cross-temporal feature interaction but also extracts features from multiple domains, offering complementary information across temporal images.
3) Finally, the custom contrastive learning model for hyperspectral binary change detection UCLM enhances the network's ability to extract features from different categories, reducing performance instability caused by data imbalance and large intraclass sample diversity.

The remainder of this article is divided into four sections. The proposed MHCD framework will be presented in Section II. Afterward, In Section III, the detailed qualitative and quantitative experimental results and analysis are discussed to verify the effectiveness of the proposed algorithm. Finally, some conclusions and further work are given in Section IV.

II. Proposed MHCD Framework

The MHCD framework aims at improving baseline HCD method results by incorporating differential band reweighting and unbalanced contrastive learning. It consists of three main components: the differential band reweighting module, the multitemporal feature interaction and fusion based on MTIM and MDFM, and the unbalanced contrastive learning based on multiple prototypes. Fig. 1 shows its overall architecture.

In the first stage, bi-temporal images are input to the DBRN. Guided by the change detection task, the differential information of the two images is analyzed and the most discriminative bands are selected. Subsequently, the images with selected bands are set in input to the Siamese detection network, where fine-grained multitemporal features are extracted through the MTIM and MDFM modules to enhance the classification. Finally, the obtained classification features are used for learning in the UCLM to adapt to the imbalanced and significantly different data structures to the final goal of a change binary detection, while simultaneously performing classification.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:49:48 UTC from IEEE Xplore. Restrictions apply.

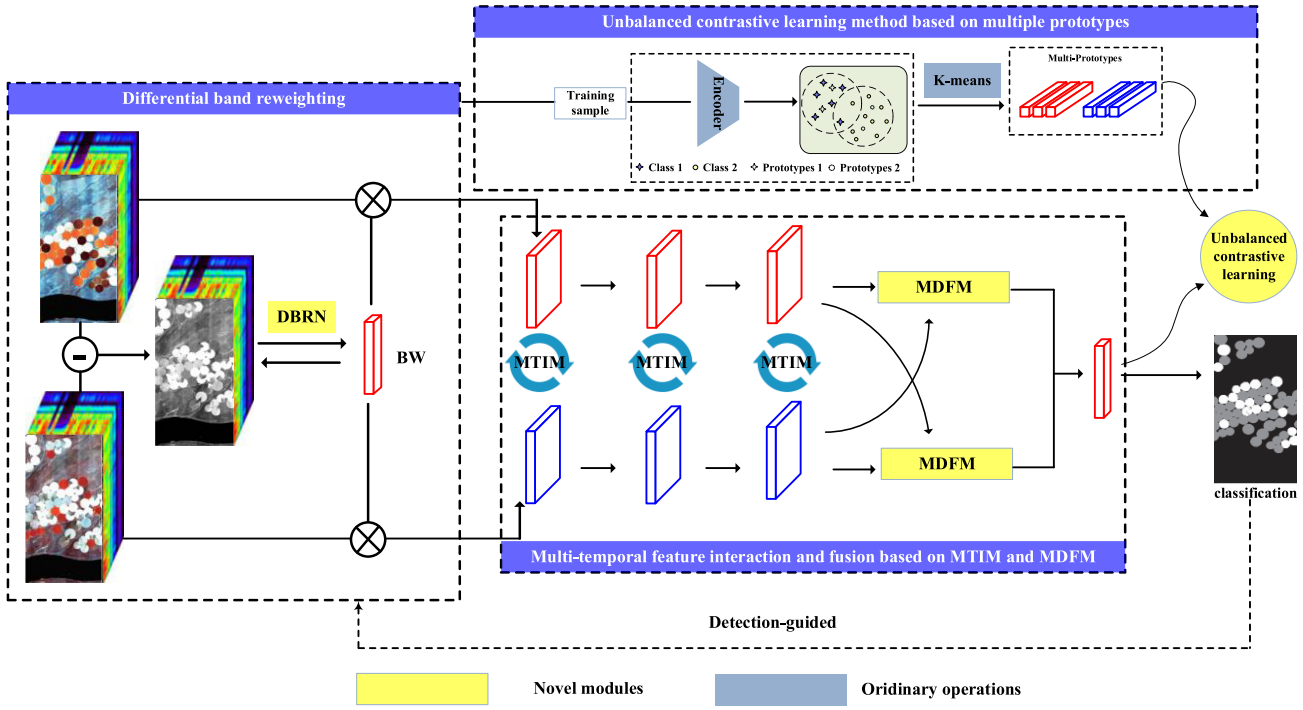


Fig. 1. Overall architecture of the proposed MHCD framework, highlighting the novel designed modules in yellow and the processing phases in blue.

A. Differential Band Reweighting

Multitask mode for band reweighting and change detection has already been considered in technical literature. However, existing approaches in this domain are characterized by one-way frameworks. In these frameworks, band reweighting is used as an initial step, and it does not engage in collaborative learning with the change detection step to adaptively optimize the overall processing chain to possibly different tasks. Furthermore, band selection techniques usually consider the correlation between bands within a single HSI. In scenarios where the input data spans multiple time instants, these conventional algorithms face challenges in capturing the changes in the time series, due to the influence of noise or outliers.

To address these problems and enhance the adaptability of band reweighting algorithms to process multitemporal (in this case change detection) data, a differential band reweighting method is proposed. The structure of DBRN is illustrated in Fig. 2. MHCD considers bi-temporal data as input with the aim of learning a set of shared sparse weights through the network. These weights are then allocated to the original HSI to select bands that are both discriminative and advantageous for the change detection task. Simultaneously, the DBRN incorporates a reconstruction module to ensure that the selected bands reconstruct as much as possible the original data, ensuring that the chosen band combination is minimally redundant. Last, the entire DBRN is end-to-end trained.

To formalize this idea, let us consider a bi-temporal set of HSIs, each consisting of b spectral bands and $H \times W$ pixels, named as $I_1 \in \mathbb{R}^{H\times W\times b}$ and I_2, respectively. Patches with a size of n extracted from I_1 and I_2 are denoted as $\{p_1^i\}_{i=1}^n$ and $\{p_2^i\}_{i=1}^n$. The labels corresponding to each sample is called y_i, and C classes are available. The band sets before and after selection are $B = \{B_i\}_{i=1}^b$ and B', respectively, where B_i is the ith band of I. The number of elements of B' is usually less than that of B. At first, two HSIs are subtracted, and the differential result is fed into the feature extraction network $\varphi_{\text{band}}(\cdot)$ composed of two convolutional layers with 3×3 convolutional kernels and a full connection layer with a sigmoid activation function. The Sigmoid function can map the values in a vector to between 0 and 1, corresponding to the weights of that band. Then the band weight (BW) ω is obtained. This process can be represented as

$$\omega = \varphi_{\text{band}}\left(p_1^i - p_2^i\right). \tag{1}$$

Afterward, to ensure that the band set B' can be reconstructed to be difference information, and that the choice of bands is sparse, the differential information is multiplied by ω and input to an AE $\varphi_{\text{recon}}(\cdot)$ with sparse and reconstruction losses to regulate the feature extraction network $\varphi_{\text{band}}(\cdot)$. The encoder–decoder consists of two convolutional layers followed by two deconvolutional layers, which is used for the reconstruction of HSIs. They all utilize the rectified linear unit (ReLU) activation function and convolutional kernels of size 3×3. The reconstructed image $\hat{p}$ is therefore obtained

$$\hat{p} = \varphi_{\text{recon}}\left(\omega \cdot \left(p_1^i - p_2^i\right)\right). \tag{2}$$

In order to make the difference between the band-selected features and the original features as small as possible, the mean-square error is selected as the loss function. In addition to the reconstruction loss, a L_1 norm constraint with a coefficient λ is employed for sparsity

$$L = \frac{1}{2n}\sum_{i=1}^{n}\left\|\hat{p} - \left(p_1^i - p_2^i\right)\right\|_2^2 + \lambda\sum_{i=1}^{n}\|\omega_i\|_1. \tag{3}$$

To be sure that change detection directly guides the DBRN updates, this loss is supplemented with a change detection classification loss, modifying it as

$$L_1 = \frac{1}{2n}\sum_{i=1}^{n}\left\|\hat{p} - \left(p_1^i - p_2^i\right)\right\|_2^2 + \lambda\sum_{i=1}^{n}\|\omega_i\|_1 - \sum_{i=1}^{n} p\log(q) + (1-p)\log(1-q)] \tag{4}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:49:48 UTC from IEEE Xplore. Restrictions apply.

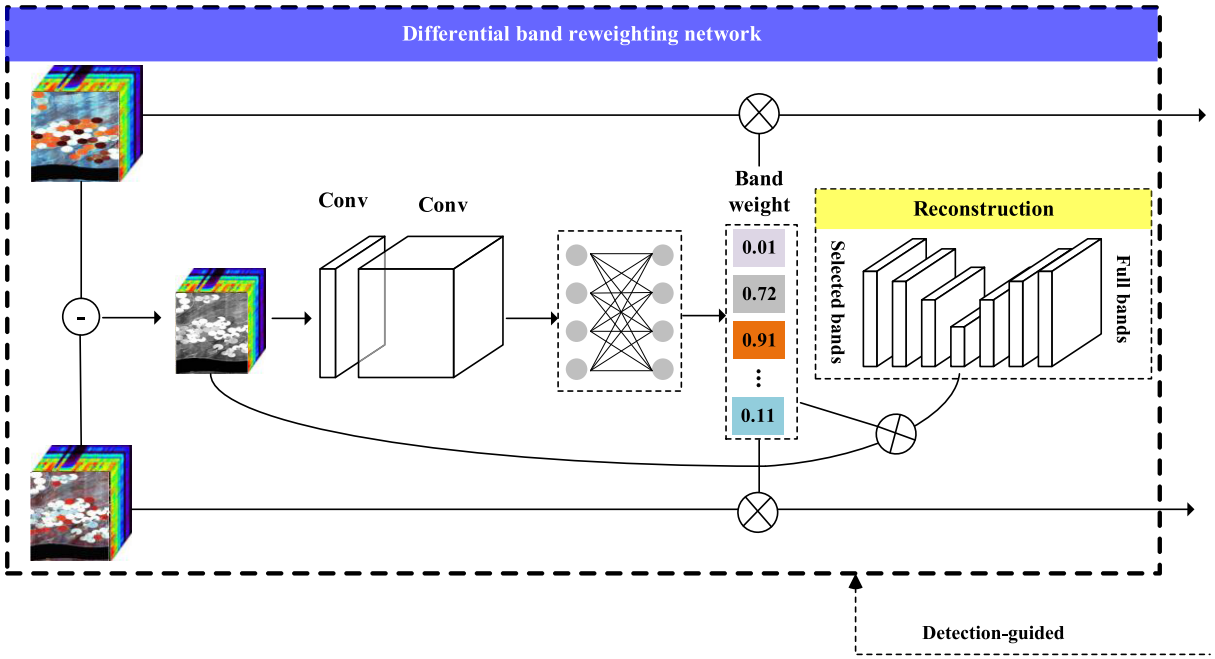


Fig. 2. Architecture of the differential band selection processing phase.

where p and q are the probability of the true label and the probability predicted by the model, respectively. Finally, the new HSIs after band reweighting are obtained through the same ω

$$\tilde{p}_1^i = p_1^i \cdot \omega$$
$$\tilde{p}_2^i = p_2^i \cdot \omega. \tag{5}$$

Being a specialized band reweighting network tailored for change detection, DBRN can be guided by task-specific loss to extract differential information across different temporal phases. This process involves the selection of an informative subset of bands, optimized for downstream tasks. The optimization strategy facilitates seamless integration of DBRN into the HCD framework, enabling bidirectional optimization for both tasks.

B. Multitemporal Feature Interaction and Fusion Based on MTIM and MDFM

The interaction among features is a crucial aspect of change detection. On the one side, multitemporal images depict features from the same scene, so they are often highly correlated, and the extracted features contain a lot of complementary information. On the other side, the different acquisition times ensure that HSIs differ in noise, shadow, and sensor parameters. This allows the consideration of two HSIs to be viewed as coming from two different domains. Thus, through feature exchange, the feature distributions of the HSIs become more similar, achieving automatic domain adaptation between the two temporal domains to a certain extent.

To enhance the interaction of temporal features for better performances, two modules specifically designed for change detection, the MTIM and the MDFM, are proposed and embedded into a backbone Siamese network. The MTIM structure is illustrated in Fig. 3, and it is integrated into the backbone network from shallow to deep layers. MTIM takes as inputs two temporal features and adaptively learns two 2-D

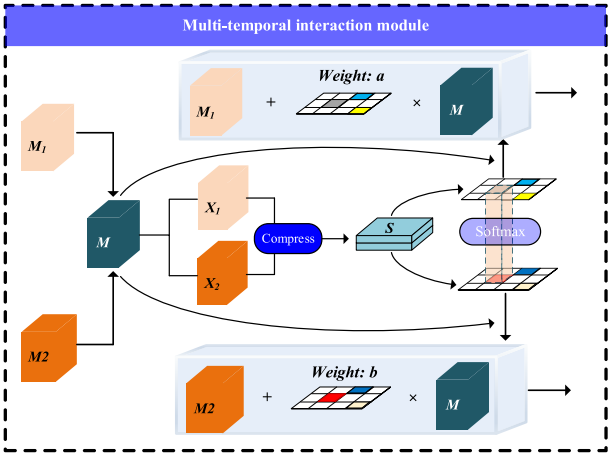


Fig. 3. Schematic of the MTIM. Features at the same instant of time (whether t1 or t2) are depicted with the same color, while dark green denotes fused features.

spatial weights through the aggregation and recombination of features. With these weights, the differential information of the two temporal features can be selectively injected into another branch. As for MDFM, it is embedded at the end of the network, as depicted in Fig. 4. MDFM achieves multidomain perception by generating spatial, spectral, and temporal attention maps. They are dynamically fused by a set of generated weights to show different attention to these three domains.

Specifically, after inputting the samples $\tilde{p}_1^i$ and $\tilde{p}_2^i$ to the basic Siamese network, the two M_1^i and M_2^i. features for the ith layers. MTIM adds M_1^i and M_2^i via element-wise summation $M = M_1^i + M_2^i$. Then, M is processed by two convolutional layers with different kernels of size 3×3, obtaining two groups of features, X_1^i and X_2^i. The compression operation consists of two convolutional layers. The compress operation maps $X_1^i + X_2^i$ to a matrix s with two channels. Based on the s, two weight maps a and b are generated by

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:49:48 UTC from IEEE Xplore. Restrictions apply.

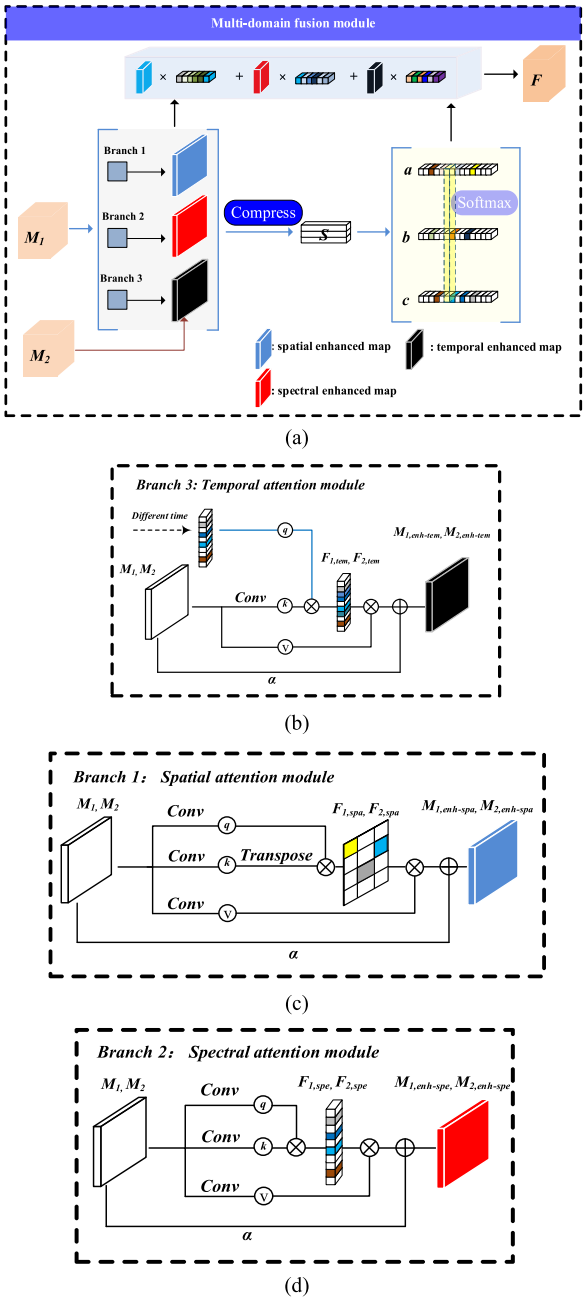


Fig. 4. Schematics of (a) MDFM and (b)–(d) spatial, spectral, and temporal attention modules.

Softmax normalization along the spectral domain

$$a(i,j)=\frac{e^{a_j s_i}}{e^{b_j s_i}+e^{b_j s_i}}, b(i,j)=\frac{e^{b_j s_i}}{e^{b_j s_i}+e^{b_j s_i}}$$
$$a(i,j)+b(i,j)=1 \tag{6}$$

where a_j and s_i are the jth column of a and ith row of s. The generated weight maps allow MTIM to select specific information from the aggregated feature M to be added to the original features, an operation denoted as $M_1 \leftarrow M_1 + a \cdot M$ and $M_2 \leftarrow M_2 + b \cdot M$. MTIM performs the same operation at each layer of the Siamese backbone network. This consistent operation enhances the flexibility of feature learning and promotes more frequent cross-temporal and cross-spatial interactions within the network.

The other module (MDFM) is embedded at the end of the backbone to extract and fuse features from the spatial, spectral, and temporal domains. The M_1 and M_2 feature are set in input to spatial, spectral, and temporal attention modules to generated $F_{1,\text{spa}}$, $F_{2,\text{spa}}$, $F_{1,\text{spe}}$, $F_{2,\text{spe}}$, $F_{1,\text{temp}}$, and $F_{2,\text{temp}}$, respectively. In the spatial attention module, convolutional layers are applied to M_1 and M_2 to get the query, key, and value features. We calculate the similarity between the query and the key and then sum these values weighted by the weights to obtain the final attention representation. The attention feature F_{spa} can be obtained

$$F_{1,\text{spa}}=\text{softmax}\left(\frac{F_{1,q}F_{1,k}^{\text{Trans}}}{\sqrt{C}}\right)$$
$$F_{2,\text{spa}}=\text{softmax}\left(\frac{F_{2,q}F_{2,k}^{\text{Trans}}}{\sqrt{C}}\right) \tag{7}$$

where $F_{1,q}$ and $F_{1,k}$ are query and key features of M_1, and C is the dimension of $F_{1,q}$. $F_{1,v}$ and $F_{2,v}$ are multiplied by these attentions and the corresponding spatial enhanced maps are $M_{1,\text{enh}-\text{spa}} = M_1 + F_{1,\text{spa}}F_{1,v}$ and $M_{2,\text{enh}-\text{spa}} = M_2 + F_{2,\text{spa}}F_{2,v}$. Similarly, the spectral enhanced maps are acquired by $M_{1,\text{enh}-\text{spc}} = M_1 + F_{1,\text{spc}}F_{1,v}$ and $M_{2,\text{enh}-\text{spc}} = M_2 + F_{2,\text{spc}}F_{2,v}$.

In the temporal attention module, there is a difference: the query features are the spectral attention maps from the other branch to perform complementary temporal feature fusion. The main differences between multitemporal images lie in changes in land cover, variations in atmospheric conditions, and differences in illumination conditions. These changes lead to differences in both spatial and spectral features among the images taken at different times. However, spatial features are highly abstracted in the deep layers of the network, making it prone to instability when constructing spatial attention. Therefore, in designing the temporal attention module, spectral information interaction is utilized among multitemporal features. Such an operation allows temporal attention maps to be generated with reference to another time. Thus, the temporal enhanced map can be formulized

$$M_{1,\text{enh}-\text{tem}} = M_1 + F_{2,\text{spc}}M_1$$
$$M_{2,\text{enh}-\text{tem}} = M_2 + F_{1,\text{spc}}M_2. \tag{8}$$

Eventually, the spatial, spectral, and temporal enhanced maps from the three different domains are integrated. Considering for instance the features from one branch, $M_{1,\text{enh}-\text{spa}}$, $M_{1,\text{enh}-\text{spc}}$, and $M_{1,\text{enh}-\text{tem}}$ are first processed by a convolutional layer and then added $M_{1,\text{mid}} = M_{1,\text{enh}-\text{spa}} + M_{1,\text{enh}-\text{spc}} + M_{1,\text{enh}-\text{tem}}$ to aggregate information from different domains. Subsequently, a compression operation with two convolutional layers and a global average pooling layer is applied to $M_{1,\text{mid}}$, resulting in a matrix $q \in \mathbb{R}^{C\times 3}$. Based on the matrix q, three weights q_1, q_2, and q_3 are calculated by the Softmax normalization along the second dimension of s. Eventually, the spatial, spectral, and temporal enhanced maps are weighted and added

$$M_{1,\text{fuse}} = q_1 \cdot M_{1,\text{enh}-\text{spa}} + q_2 \cdot M_{1,\text{enh}-\text{spc}} + q_3 \cdot M_{1,\text{enh}-\text{tem}}. \tag{9}$$

After $M_{2,\text{fuse}}$ is obtained in the same way $z = M_{1,\text{fuse}} + M_{2,\text{fuse}}$ is used for classification.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:49:48 UTC from IEEE Xplore. Restrictions apply.

C. Unbalanced Contrastive Learning Based on Multiple Prototypes

In most change detection problems, change pixels are far fewer than no change ones. This imbalance may lead to imbalanced training sets. Such imbalance may cause the model to overfit a particular class and shift the decision boundaries, ultimately affecting the model's performance and generalization ability. Additionally, in HCD, change and no-change pixels equally belong to extremely diverse spectral classes. Therefore, there is significant intraclass variation, making it difficult to obtain sharp classification boundaries.

To address this issue, we propose a novel UCLM embedded into the MHCD framework. The UCLM leverages the fused features z in output to MDFM as contrasting features. On the one hand, the UCLM estimates multiple prototypes for each class based on the features of all training samples before each forward inference. For each sample feature, they look for the closest class prototype for loss calculation. On the other hand, UCML weakens the influence of the class with a larger number of samples by averaging the contributions of same-class samples to the contrastive loss during the loss calculation.

First, the original prototypical contrastive learning [39] is introduced. In a batch B with N samples, for a feature of instance z_i, the loss has following expression:

$$L_p = -\sum_{i=1}^{N}\left(\log\frac{\exp\left(z_i\cdot z_i'/\tau\right)}{\sum_{j=0}^{r}\exp\left(z_i\cdot z_j'/\tau\right)} + \frac{1}{M}\sum_{m=1}^{M}\log\frac{\exp\left(z_i\cdot c_s^m/\phi_s^m\right)}{\sum_{j=0}^{r}\exp\left(z_i\cdot c_j^m/\phi_j^m\right)}\right) \tag{10}$$

where c_s is the prototype. r refers to the number of negative features and τ is a hyperparameter to control tolerance to similar samples. ϕ_s is the concentration estimation and r is the number of negative samples corresponding to z_i. This is an unsupervised prototype-contrast loss, where clustering is performed M times with different numbers of clusters. The second term in the formula represents multiple estimations of class prototypes when labels are unavailable. The first term acts as the normalization term and the second term as prototypical contrastive learning.

First, data augmentation is performed by random cropping, random horizontal flipping, and adding Gaussian noise. UCLM train MHCD follows the pattern of original prototypical learning [39]. In the UCLM, after updating MHCD, a momentum encoder with the same structure as MHCD is constructed to infer the output feature set of all training samples and augmentation samples $\{z_i\}_{i=1}^{n}$. The momentum encoder network parameters are directly updated by the following formula:

$$\theta_k \leftarrow m\theta_k + (1-m)\theta_q \tag{11}$$

where θ_q and θ_k is the parameter of MHCD and momentum encoder network. And m is a factor between 0 and 1. For each class, multiple prototypes are estimated using the k-means method. Specifically, the features $\{z_i\}_{i=1}^{n}$ are divided into two groups according to the class, $\{z_i^1\}_{i=1}^{m}$ and $\{z_i^2\}_{i=1}^{n}$. Then, the k-means algorithm is applied to each group to obtain prototypes. Taking the features of the first class $\{z_i^1\}_{i=1}^{m}$ as an example, K samples are randomly selected as initial class centers $\{c_1^1,\ldots,c_K^1\}$. For each feature z_i^1, its distance to the class centers is calculated, and the feature is assigned to the class corresponding to the minimum distance. Then, for each class center, the cluster center is recalculated. Repeat these steps until convergence and two prototype sets for two classes $\{c_1^1,\ldots,c_k^1\}$ and $\{c_1^2,\ldots,c_k^2\}$ are obtained. For each feature z_i, find its nearest prototypes in the corresponding prototype set

$$k^* = \arg\min\|z_i - c_k\|(k=1,\ldots K) \tag{12}$$

where k^* is the subscript of prototype and $\|\cdot\|$ denotes the Euclidean distance. For each sample feature, they find the closest prototype for calculating the contrastive loss. Accordingly, the contrastive loss with multiple prototypes is modified as

$$L_p = -\sum_{i=1}^{N}\left(\log\frac{\exp\left(z_i\cdot z_i'/\tau\right)}{\sum_{j=0}^{r}\exp\left(z_i\cdot z_j'/\tau\right)} + \log\frac{\exp\left(z_i\cdot c_s/\phi_s^k\right)}{\sum_{j=0}^{r}\exp\left(z_i\cdot c_j/\phi_j^k\right)}\right). \tag{13}$$

Since MHCD has access to labels, there is no need to estimate class prototypes multiple M times. Instead, it is necessary to determine the nearest prototype corresponding to each feature. The first term is a normalization that uses noise-contrastive estimation (NCE). The second term is supervised prototypical contrastive learning with multiple prototypes.

Considering the significant intraclass variation, it is necessary to average the contributions of instances from different classes within a batch for optimization. We introduce an averaging operation to balance, which takes the following form:

$$L_2 = -\sum_{i=1}^{N}\left(\log\frac{\exp\left(z_i\cdot z_i'/\tau\right)}{\sum_{j=0}^{r}\exp\left(z_i\cdot z_j'/\tau\right)} + \log\frac{\exp\left(z_i\cdot c_s/\phi_s^k\right)}{\sum_{m=0}^{C}\frac{1}{|B_m|}\sum_{j=0}^{r}\exp\left(z_i\cdot c_j/\phi_j^k\right)}\right) \tag{14}$$

where $|B_m|$ is the number of samples of class m. In this case, the loss is averaged in front of the exponential function, resulting in a reduction of the dominant influence of head classes and an emphasis on the contribution of classes with fewer samples.

III. Experimental Results and Analysis

A. Datasets

To evaluate the effectiveness of the proposed method, experiments on three different datasets were performed. The information about the sensor, location, and size of these three datasets is summarized in Table I. They cover different environments to allow a more comprehensive validation of the performances of the proposed approach under various imaging conditions. The Hermiston dataset, in particular, has a positive-to-negative sample ratio of 1:6.8, which can validate the proposed algorithm's efficacy in handling strongly imbalanced annotated datasets.

B. Evaluation Metrics

In the experiments, both quantitative and qualitative analyses were performed. For qualitative analysis, visualizing the classification results allows for an intuitive observation of the effectiveness of the proposed change detection technique. For quantitative analysis, several commonly used metrics were

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:49:48 UTC from IEEE Xplore. Restrictions apply.

TABLE I
DETAILED INFORMATION OF THREE DATASETS

Name	Sensor	Location	Size	Captured time
Farmland	EO-1	Jiangsu, China	420×140×154	2006-2007
BayArea	AVIRIS	Pattersoncity, USA	600×500×224	2013-2015
Hermiston	Hyperion	Hermiston city, USA	390×200×242	2004-2007

TABLE II
DEFINITIONS AND DESCRIPTIONS OF THE METRICS

Metric	Statements	Optimal value	Definition
OA	Total correct predictions	1	$OA = \frac{TP+TN}{TP+FP+TN+FN}$
Kappa	Agreement between predicted and reference	1	$Kappa = \frac{OA-Pe}{1-Pe}$, $Pe = \frac{(TP+FP)(TP+FN)+(FN+TN)(FP+TN)}{(TP+TN+FP+FN)^2}$
Recall	Ability of identifying positive instances	1	$\mathrm{Re} = \frac{TP}{TP+FN}$
Precision	The accuracy of the positive predictions	1	$\mathrm{Pr} = \frac{TP}{TP+FP}$
F1	A balanced measure that considers both false positives and false negatives	1	$F1 = 2 \cdot \frac{\mathrm{Pr} \cdot \mathrm{Re}}{\mathrm{Pr}+\mathrm{Re}}$

selected to evaluate the performances, i.e., overall accuracy (OA), Kappa coefficient (Kappa), recall, precision, and F1 score (F1). These metrics comprehensively reflect the model's classification performance for positive and negative samples. Recall measures the proportion of actual positive cases that were correctly identified by the model, precision measures the proportion of predicted positive cases that were truly positive. The F1 score balances both precision and recall, providing a single metric that summarizes the model's performance. The computation of these five metrics is based on the true positive (TP), true negative (TN), false positive (FP), and false negative (FN) values in the confusion matrix. The specific calculation methods and descriptions are provided in Table II.

C. Experimental Set-Up

To verify the effectiveness of the algorithm, various state-of-the-art methods, both traditional and deep learning-based, were used for a comparative analysis. Traditional methods included the most commonly used PCA [12] and CVA [8]. Additionally, four novel and recent deep learning-based methods were selected, including two-dimensional convolutional neural network (2DCNN) [40], bilinear CNNs (BCNNs) [41], cross-band 2-D attention network (CBANet) [28], and multiscale diff-changed feature fusion network (MSDFFN) [42]. For 2DCNN, BCNNs, CBANet, and MSDFFN, their optimal hyperparameters such as batch size and learning rate, were determined through trial and error. For MHCD, the batch size, learning rate, and number of epochs are set to 64, 0.0001, and 300, respectively. For methods requiring input with a spatial window, whose size was uniformly set to 9×9 pixels. All deep learning methods were implemented using PyTorch and experiments were performed on an NVIDIA RTX3080 GPU. For each dataset, 5% of labeled samples from each category are selected as training samples, with the remaining samples used for testing.

D. Detection Results

The quantitative results of the proposed MHCD method and the comparative methods are reported in the following tables, with the best result for each metric highlighted in bold. Additionally, the classification results are visualized in subsequent images to provide a more intuitive representation of the method's strengths and weaknesses in terms of edge and region consistency. We have emphasized areas of significant visual difference with red boxes to make it easier for the reader to recognize what is mentioned in the text in the spatial context of the scene.

1) Farmland Dataset: Table III presents the change detection results of the compared seven methods. Based on the OA results, it can be observed that most deep learning algorithms are capable of distinguishing changed and unchanged pixels, though the differences among them are not particularly significant. However, looking at the F1 score, it becomes evident that deep learning algorithms exhibit a greater advantage in classification accuracy across different categories. Although CBANet achieves the best precision, it misclassifies many changed pixels as unchanged, resulting in a decrease in recall. MHCD achieves the best performance across most metrics and strikes a good balance between precision and recall, leading to accurate change detection results on this dataset.

Fig. 5 visually illustrates the classification results. PCA and CVA tend to misclassify many unchanged pixels as changed pixels. Deep learning algorithms such as 2DCNN, BCNNs, CBANet, and MHCD demonstrate superior region consistency. However, compared to MHCD, other deep learning algorithms exhibit excessive spatial smoothing. Particularly, in the top-left corner of the image, MHCD achieves the best classification performance along the edges.

2) BayArea Dataset: The change detection results of the BayArea dataset are shown in Table IV. The characteristic of this dataset lies in the complex distribution of changing

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:49:48 UTC from IEEE Xplore. Restrictions apply.

TABLE III
CHANGE DETECTION RESULTS ON THE FARMLAND DATASET

Method	OA	AA	KAPPA	Precision	Recall	F1
PCA	96.08	95.51	90.54	94.16	92.47	93.30
CVA	95.23	94.78	88.55	93.70	90.24	91.94
2DCNN	95.59	95.65	91.76	94.98	93.37	94.17
BCNNs	96.66	95.83	91.92	94.76	93.80	94.29
CBANet	96.68	95.43	92.05	**96.69**	92.23	94.41
MSDFFN	97.17	96.32	93.19	96.26	94.15	95.19
MHCD	**97.31**	**96.68**	**93.49**	95.63	**95.14**	**95.39**

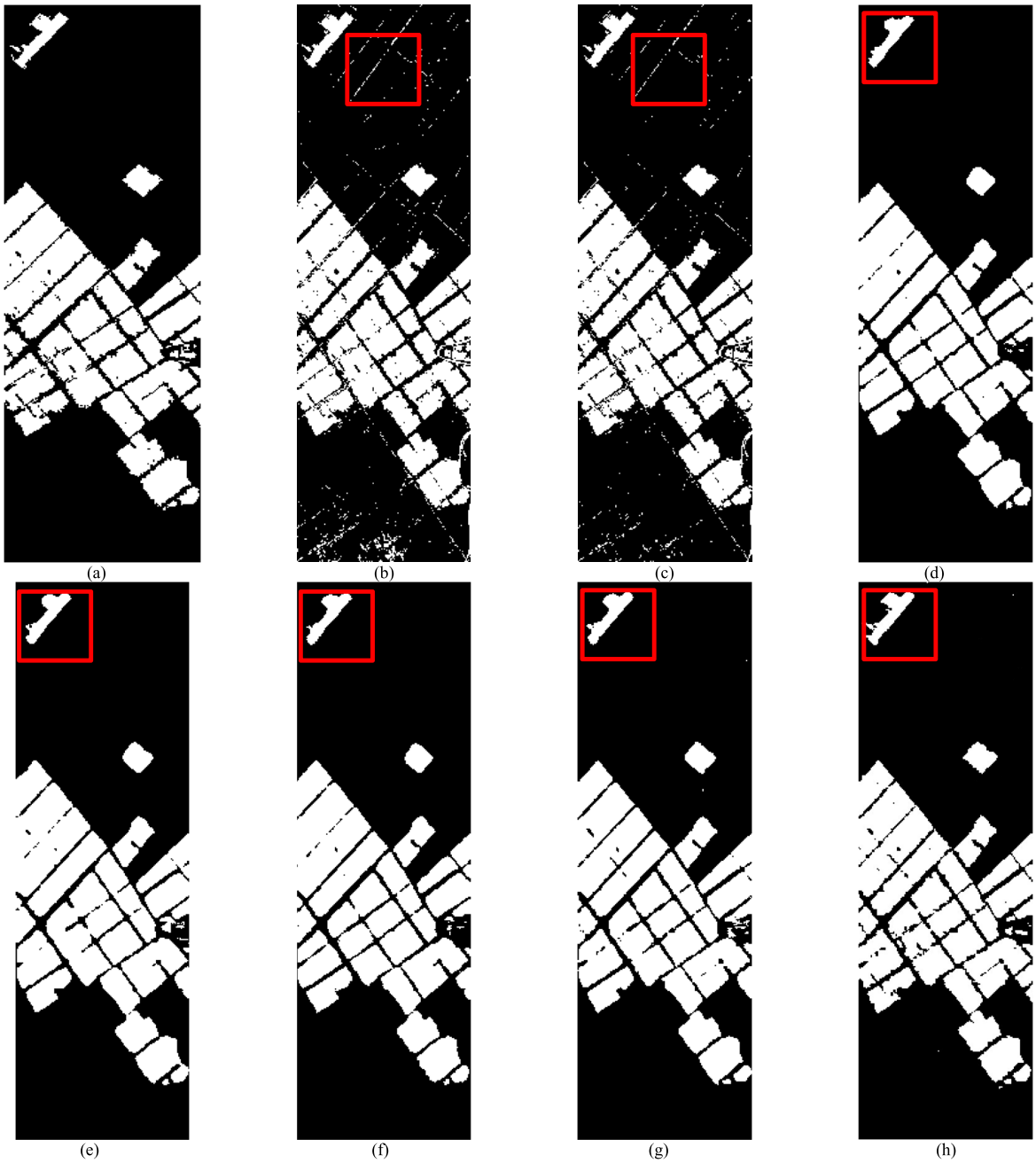


Fig. 5. Visualization of the change detection maps for different methods on the Farmland datasets. The areas in red are mentioned in the text. (a) GT. (b) PCA. (c) CVA. (d) 2DCNN. (e) BCNNs. (f) CBANet. (g) MSDFFN. (h) MHCD.

regions, making it more challenging to accurately detect changes. In terms of OA, traditional algorithms exhibit numerous FPs in complex scenarios. Compared to PCA and CVA, the deep learning algorithms show significant

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:49:48 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
CHANGE DETECTION RESULTS ON THE BAYAREA DATASET

Method	OA	AA	KAPPA	Precision	Recall	F1
PCA	84.07	84.73	68.45	94.30	76.78	84.64
CVA	82.17	85.83	70.63	95.38	77.81	85.69
2DCNN	96.00	96.49	91.93	91.59	**99.81**	95.52
BCNNs	98.10	98.06	96.19	98.69	97.27	97.97
CBANet	98.25	98.19	96.49	**99.67**	96.68	98.15
MSDFFN	98.45	98.40	96.89	99.56	97.19	98.36
MHCD	**99.09**	**99.07**	**98.18**	99.34	98.72	**99.03**

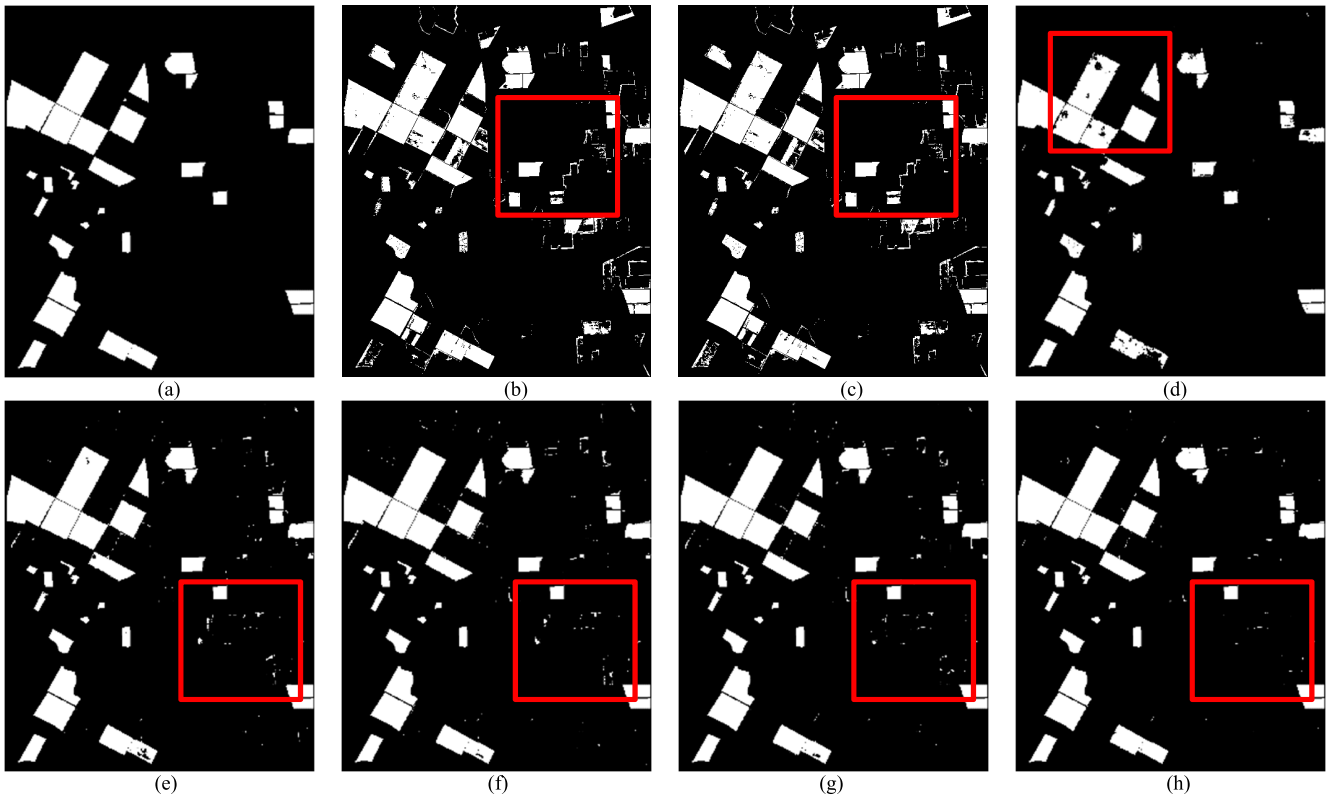


Fig. 6. Visualization of the change detection maps for different methods on the BayArea datasets. The areas in red are mentioned in the text. (a) GT. (b) PCA. (c) CVA. (d) 2DCNN. (e) BCNNs. (f) CBANet. (g) MSDFFN. (h) MHCD.

improvements in class accuracy and OA, largely due to their deep feature extraction capabilities. However, from the perspective of Precision and Recall, algorithms such as CBANet and MSDFFN exhibit more FNs and FPs. The advantages of MHCD in average accuracy (AA) and KAPPA are particularly evident, indicating that MHCD can effectively balance between the two classes.

Fig. 6 displays the visual results of change detection in the BayArea. PCA and CVA algorithms are sensitive to the boundaries of buildings, often misclassifying them as changed. 2DCNN tends to classify many changed pixels as unchanged. BCNNs, CBANet, and MSDFFN exhibit a lot of spatial noise. The results of MHCD are closest to the ground truth (GT), particularly maintaining good region consistency and containing fewer spatial noises.

3) Hermiston Dataset: Table V presents the results of the Hermiston dataset. Compared to the previous two datasets, this dataset has a relatively large number of positive and negative samples. For traditional algorithms like PCA and CVA, which do not rely on training samples, their precision and recall do not differ significantly. However, in cases of sample imbalance, precision and recall for 2DCNN, BCNNs, CBANet, and MSDFFN become extremely imbalanced. Among these six algorithms, MHCD achieves a better balance between precision and recall and obtains the best F1 score, indicating that the algorithm effectively addresses the imbalance introduced by unbalanced contrastive learning.

Fig. 7 displays the classification results of the Hermiston dataset. BCNNs, CBANet, and MSDFFN show spatial effects closest to MHCD, however, they are more prone to misclassification at the edge details. Compared to deep learning algorithms and traditional algorithms, MHCD exhibits prominent classification performance at the edges between different classes, with the least amount of noise.

E. Ablation Study

We validated the effectiveness of the proposed method's different modules through five ablation experiments on three datasets and the results of the experiments are recorded in Tables VI–VIII. We denote DBRN as M1, MTIM as M2,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:49:48 UTC from IEEE Xplore. Restrictions apply.

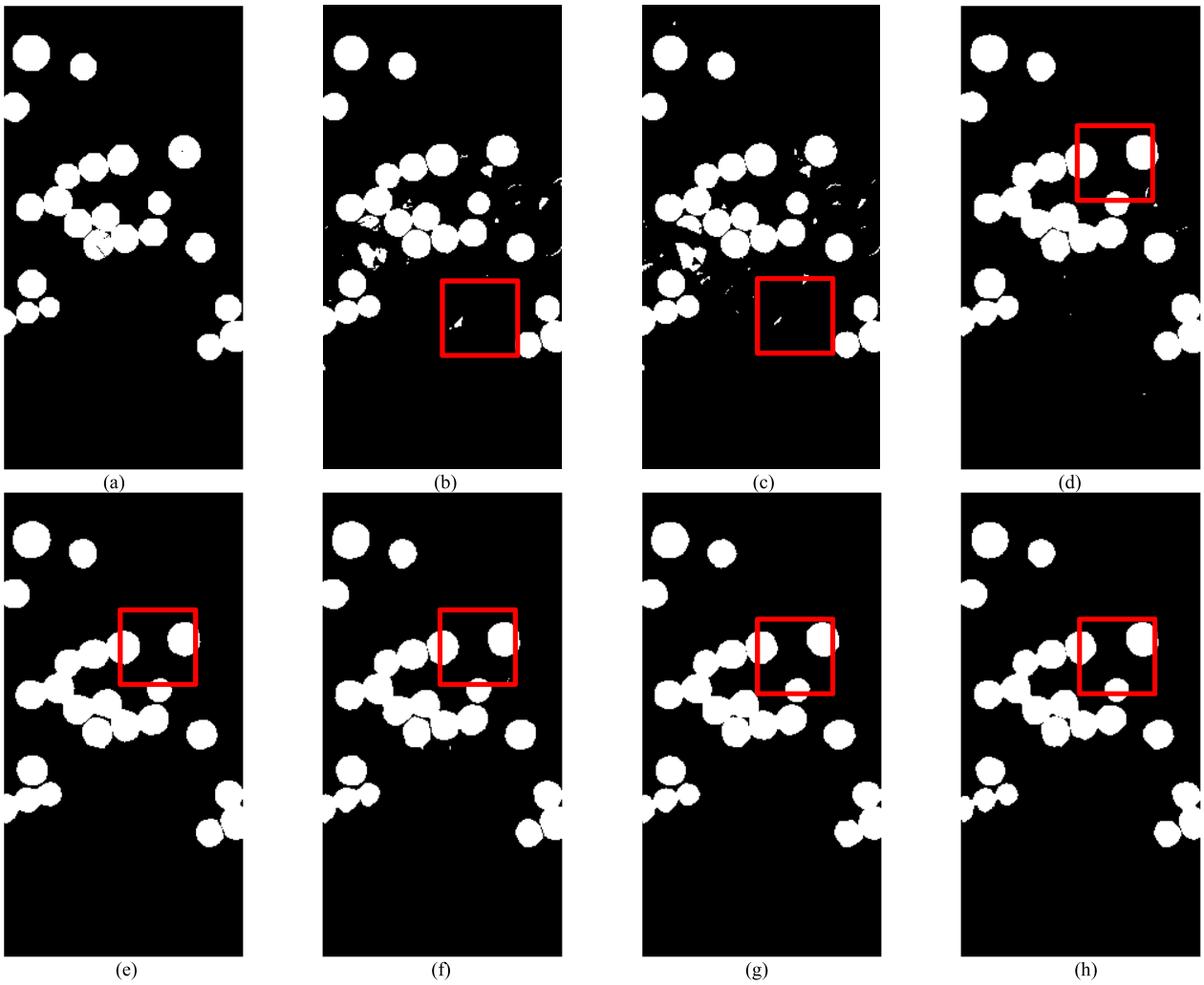


Fig. 7. Visualization of the change detection maps for different methods on the Hermiston datasets, the areas in red are mentioned in the text. (a) GT. (b) PCA. (c) CVA. (d) 2DCNN. (e) BCNNs. (f) CBANet. (g) MSDFFN. (h) MHCD.

TABLE V
CHANGE DETECTION RESULTS ON THE HERMISTON DATASET

Method	OA	AA	KAPPA	Precision	Recall	F1
PCA	97.53	95.75	89.21	93.36	88.07	90.63
CVA	97.81	95.98	90.36	93.51	89.80	91.62
2DCNN	98.42	94.67	93.25	**99.44**	89.42	94.16
BCNNs	98.46	94.91	93.37	98.97	89.98	94.26
CBANet	98.71	95.70	94.43	99.13	91.52	95.17
MSDFFN	99.03	96.72	95.76	99.26	93.55	96.32
MHCD	**99.15**	**97.60**	**96.23**	97.98	**95.50**	**96.72**

MDFM as M3, and UCLM as M4. When M1 is not adopted, the network will use full-band input; when M2 is not adopted, the network will not perform interaction; when M3 is not used, the features of the two branches will be directly summed for change detection; when M4 is not used, the network will lose additional contrastive learning loss. Five experiments use different combinations of modules. Three metrics, OA, Kappa, and F1 are selected to evaluate their performances.

As shown in Tables VI–VIII, the comparison between Experiments 1 and 5 indicates that DBRN can enhance performance by assigning higher weights to discriminative bands. The comparison between Experiments 2 and 5 shows that MTIM improves change detection performance through the interaction of multiple temporal phases, particularly achieving an increase of more than 1% in KAPPA over the Hermiston dataset. Regarding Experiments 3 and 5, they demonstrate that MDFM is effective in improving performance. Consistent performance improvements validate the effectiveness of the designed contrastive learning by comparing Experiments 4 and 5.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:49:48 UTC from IEEE Xplore. Restrictions apply.

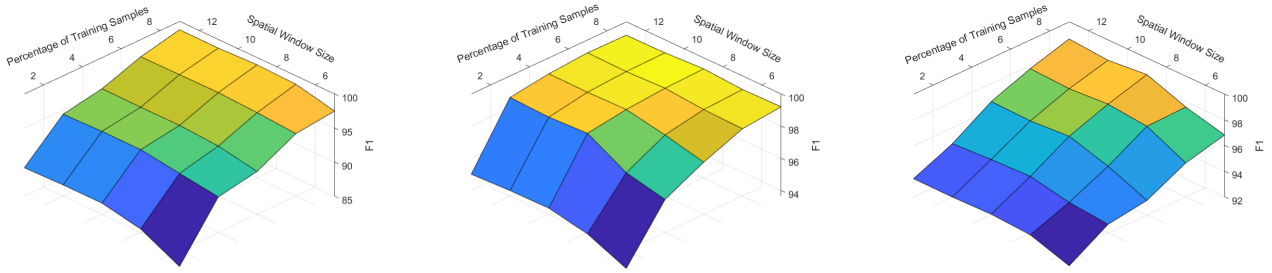


Fig. 8. Sensitivity analysis of the number of training samples and the spatial window size on different datasets. (a) Farmland. (b) BayArea. (c) Hermiston.

TABLE VI
ABLATION STUDY ON THE FARMLAND DATASET

NO	Module				Metrics		
	M1	M2	M3	M4	OA	Kappa	F1
1		√	√	√	97.02	92.74	94.84
2	√		√	√	97.15	93.19	95.22
3	√	√		√	97.21	93.34	95.33
4	√	√	√		97.27	93.36	95.28
5	√	√	√	√	**97.31**	**93.49**	**95.39**

TABLE VII
ABLATION STUDY ON THE BAYAREA DATASET

NO	Module				Metrics		
	M1	M2	M3	M4	OA	Kappa	F1
1		√	√	√	98.45	96.88	98.32
2	√		√	√	98.30	97.65	98.89
3	√	√		√	98.56	97.11	98.63
4	√	√	√		98.84	97.67	98.75
5	√	√	√	√	**99.09**	**98.18**	**99.03**

TABLE VIII
ABLATION STUDY ON THE HERMISTON DATASET

NO	Module				Metrics		
	M1	M2	M3	M4	OA	Kappa	F1
1		√	√	√	98.41	92.96	93.88
2	√		√	√	98.81	94.73	95.42
3	√	√		√	98.77	94.55	95.26
4	√	√	√		98.97	95.15	95.78
5	√	√	√	√	**99.15**	**96.23**	**96.72**

F. Sensitivity Analysis on Training Samples and Spatial Window Size

This section investigates the performance of the algorithm in the case of different training samples and spatial window sizes. We selected 1%, 3%, 5%, 7%, and 9% of the samples for training, respectively, and set the spatial window size to 5, 7, 9, 11, and 13 pixels. The performance of the experiments is measured by computing the F1 score. The results are presented in images shown in Fig. 8. As shown in the figures, as the percentage of training samples increases, the performance under different spatial windows improves, with the most significant improvement observed on the Farmland dataset. With the increase in spatial window size, the performance initially increases due to the availability of more spatial information. However, as the spatial window size continues to increase, the performance stabilizes. Overall, it can be observed that when the spatial window is set to 9 pixels, the performance of all algorithms reaches its optimum.

TABLE IX
COMPARISON OF COMPUTATIONAL COSTS OF DIFFERENT METHODS ON THREE DATASETS

Dataset	Method	Training Times(s)	Testing Time(s)
Farmland	2DCNN	45.2	8.9
	BCNNs	258.2	21.5
	CBANet	95.3	11.8
	MSDFFN	965.1	26.7
	MHCD	139.0	16.6
BayArea	2DCNN	58.2	10.1
	BCNNs	261.7	18.1
	CBANet	114.2	11.8
	MSDFFN	1025.3	29.3
	MHCD	144.6	14.9
Hermiston	2DCNN	63.2	11.0
	BCNNs	269.3	16.6
	CBANet	132.8	14.4
	MSDFFN	1080.4	30.9
	MHCD	168.6	15.4

TABLE X
COMPARISON OF PARAMETERS OF DIFFERENT DL-BASED METHODS

Method	Parameters (K)
2DCNN	198.2
BCNNs	4542.7
CBANet	321.4
MSDFFN	39452.4
MHCD	1475.2

G. Study on the Number of Parameters and Computational Cost

In this chapter, the results of the training and testing times of deep learning algorithms, as well as the model parameter count, are summarized in Tables IX and X. The training and testing times of deep learning are primarily influenced by the size of the model. Combining the two tables, it can be observed that the training and testing times are positively correlated with the parameter count. Specifically, 2DCNN has a simpler model, resulting in shorter training time. However, MSDFFN, due to its inclusion of numerous multiscale feature extraction modules, results in a larger model size and longer training and testing times. For the MHCD algorithm, it has fewer parameters than MSDFFN, yet it achieves better results.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:49:48 UTC from IEEE Xplore. Restrictions apply.

IV. Conclusion

This article aims at developing a multitask learning framework for HCD and band reweighting. The proposed MHCD algorithm achieves mutual learning between a task-adaptive DBRN and the change detection task. Specifically, the change detection task guides the DBRN in selecting discriminative bands, while the DBRN enhances the detection performance for the changes.

Additionally, the two modules proposed in this article, MTIM and MDFM, extract features from multiple domains and have been proven to effectively enhance performance. It is worth noting that, since they are transferable modules, they can be easily incorporated into any network.

Last, we propose an UCLM suitable for change detection. It effectively addresses issues such as large intraclass differences and imbalanced distributions of labeled samples by estimating multiple prototypes and dynamically adjusting the contribution of different classes to the loss. In the future, we will explore multitask learning patterns for change detection and other visual tasks.

References

[1] Q. Huang, C. Wang, Y. Meng, J. Chen, and A. Yue, "Landslide monitoring using change detection in multitemporal optical imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 312–316, Feb. 2020.

[2] T. T. Lê, J.-L. Froger, and D. Ho Tong Minh, "Multiscale framework for rapid change analysis from SAR image time series: Case study of flood monitoring in the central coast regions of Vietnam," *Remote Sens. Environ.*, vol. 269, Feb. 2022, Art. no. 112837.

[3] C. Cao, S. Dragić ević, and S. Li, "Land-use change detection with convolutional neural network methods," *Environments*, vol. 6, no. 2, p. 25, Feb. 2019.

[4] V. Alhassan, C. Henry, S. Ramanna, and C. Storie, "A deep learning framework for land-use/land-cover mapping and analysis using multispectral satellite imagery," *Neural Comput. Appl.*, vol. 32, no. 12, pp. 8529–8544, Jun. 2020.

[5] Z. Wei, X. Gu, Q. Sun, X. Hu, and Y. Gao, "Analysis of the spatial and temporal pattern of changes in abandoned farmland based on long time series of remote sensing data," *Remote Sens.*, vol. 13, no. 13, p. 2549, Jun. 2021.

[6] G. Sofia, J.-S. Bailly, N. Chehata, P. Tarolli, and F. Levavasseur, "Comparison of Pleiades and LiDAR digital elevation models for terraces detection in farmlands," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 4, pp. 1567–1576, Apr. 2016.

[7] M. Hasanlou and S. T. Seydi, "Hyperspectral change detection: An experimental comparative study," *Int. J. Remote Sens.*, vol. 39, no. 20, pp. 7029–7083, Oct. 2018.

[8] W. A. Malila, "Change vector analysis: An approach for detecting forest changes with Landsat," in *Proc. LARS Symposia*, 1980, p. 385.

[9] F. Bovolo, S. Marchesi, and L. Bruzzone, "A framework for automatic and unsupervised detection of multiple changes in multitemporal images," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 6, pp. 2196–2212, Jun. 2012.

[10] C. Yang, J. H. Everitt, and J. M. Bradford, "Yield estimation from hyperspectral imagery using spectral angle mapper (SAM)," *Trans. ASABE*, vol. 51, no. 2, pp. 729–737, 2008.

[11] C. Wu, B. Du, and L. Zhang, "Slow feature analysis for change detection in multispectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 5, pp. 2858–2874, May 2014.

[12] J. S. Deng, K. Wang, Y. H. Deng, and G. J. Qi, "PCA-based land-use change detection and analysis using multitemporal and multisensor satellite data," *Int. J. Remote Sens.*, vol. 29, no. 16, pp. 4823–4838, Aug. 2008.

[13] A. A. Nielsen, K. Conradsen, and J. J. Simpson, "Multivariate alteration detection (MAD) and MAF postprocessing in multispectral, bitemporal image data: New approaches to change detection studies," *Remote Sens. Environ.*, vol. 64, no. 1, pp. 1–19, 1998.

[14] W. Sun et al., "A label similarity probability filter for hyperspectral image postclassification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 6897–6905, 2021.

[15] F. Bovolo, L. Bruzzone, and M. Marconcini, "A novel approach to unsupervised change detection based on a semisupervised SVM and a similarity measure," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 7, pp. 2070–2082, Jul. 2008.

[16] F. Huang, Y. Yu, and T. Feng, "Hyperspectral remote sensing image change detection based on tensor and deep learning," *J. Vis. Commun. Image Represent.*, vol. 58, pp. 233–244, Jan. 2019.

[17] F. Huang, C. Xu, and Z. Zhu, "Change detection of hyperspectral remote sensing images based on deep belief network," *Int. J. Earth Sci. Eng.*, vol. 9, pp. 2096–2105, Oct. 2016.

[18] M. Hu, C. Wu, L. Zhang, and B. Du, "Hyperspectral anomaly change detection based on autoencoder," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 3750–3762, 2021.

[19] Y. Zhang et al., "Depthwise separable convolutional autoencoders for hyperspectral image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[20] J. Li, X. Yuan, J. Li, G. Huang, L. Feng, and J. Zhang, "MV-CDN: Multi-visual collaborative deep network for change detection of double-temporal hyperspectral images," *Remote Sens.*, vol. 15, no. 11, p. 2834, May 2023.

[21] H. Ge, Y. Tang, Z. Bi, T. Zhan, Y. Xu, and A. Song, "MMSRC: A multidirection multiscale spectral–spatial residual network for hyperspectral multiclass change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 9254–9265, 2022.

[22] Q. Wang, Z. Yuan, Q. Du, and X. Li, "GETNET: A general end-to-end 2-D CNN framework for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 3–13, Jan. 2018.

[23] X. Wang, S. Li, X. Zhao, and K. Zhao, "BiG-FSLF: A cross heterogeneous domain few-shot learning framework based on bidirectional generation for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, doi: 10.1109/TGRS.2023.3292249.

[24] A. Song, J. Choi, Y. Han, and Y. Kim, "Change detection in hyperspectral images using recurrent 3D fully convolutional networks," *Remote Sens.*, vol. 10, no. 11, p. 1827, Nov. 2018.

[25] C. Shi, Z. Zhang, W. Zhang, C. Zhang, and Q. Xu, "Learning multiscale temporal–spatial–spectral features via a multipath convolutional LSTM neural network for change detection with hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, doi: 10.1109/TGRS.2022.3176642.

[26] L. Mou, L. Bruzzone, and X. X. Zhu, "Learning spectral–spatial–temporal features via a recurrent convolutional neural network for change detection in multispectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 924–935, Feb. 2019.

[27] W. Dong et al., "Abundance matrix correlation analysis network based on hierarchical multihead self-cross-hybrid attention for hyperspectral change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, doi: 10.1109/TGRS.2023.3235401.

[28] Y. Li et al., "CBANet: An end-to-end cross-band 2-D attention network for hyperspectral change detection in remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, doi: 10.1109/TGRS.2023.3276589.

[29] X. Ou, L. Liu, B. Tu, L. Qing, G. Zhang, and Z. Liang, "CBW-MSSANet: A CNN framework with compact band weighting and multiscale spatial attention for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, doi: 10.1109/TGRS.2023.3263563.

[30] Y. Wang et al., "Spectral–spatial–temporal transformers for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, doi: 10.1109/TGRS.2022.3203075.

[31] Y. Wang, J. Sha, L. Gao, Y. Zhang, X. Rong, and C. Zhang, "A semi-supervised domain alignment transformer for hyperspectral images change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, doi: 10.1109/TGRS.2023.3317919.

[32] J. Qu, S. Hou, W. Dong, Y. Li, and W. Xie, "A multilevel encoder–decoder attention network for change detection in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, doi: 10.1109/TGRS.2021.3130122.

[33] L. Wang, L. Wang, Q. Wang, and P. M. Atkinson, "SSA-SiamNet: Spectral–spatial-wise attention-based Siamese network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, doi: 10.1109/TGRS.2021.3095899.

[34] B. Yang, L. Qin, J. Liu, and X. Liu, "UTRNet: An unsupervised time-distance-guided convolutional recurrent network for change detection in irregularly collected images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, doi: 10.1109/TGRS.2022.3174009.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:49:48 UTC from IEEE Xplore. Restrictions apply.

[35] S. Saha, L. Kondmann, Q. Song, and X. X. Zhu, "Change detection in hyperdimensional images using untrained models," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 11029–11041, 2021.
[36] B. Yang, Y. Mao, L. Liu, X. Liu, Y. Ma, and J. Li, "From trained to untrained: A novel change detection framework using randomly initialized models with spatial–channel augmentation for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, doi: 10.1109/TGRS.2023.3262928.
[37] S. Liu, Q. Du, X. Tong, A. Samat, H. Pan, and X. Ma, "Band selection-based dimensionality reduction for change detection in multi-temporal hyperspectral images," *Remote Sens.*, vol. 9, no. 10, p. 1008, Sep. 2017.
[38] X. Ou, L. Liu, B. Tu, G. Zhang, and Z. Xu, "A CNN framework with slow-fast band selection and feature fusion grouping for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5524716.
[39] J. Li, P. Zhou, C. Xiong, and S. C. H. Hoi, "Prototypical contrastive learning of unsupervised representations," 2020, *arXiv:2005.04966*.
[40] N. He et al., "Feature extraction with multiscale covariance maps for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 755–769, Feb. 2019.
[41] Y. Lin, S. Li, L. Fang, and P. Ghamisi, "Multispectral change detection with bilinear convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 10, pp. 1757–1761, Oct. 2020.
[42] F. Luo, T. Zhou, J. Liu, T. Guo, X. Gong, and J. Ren, "Multiscale diff-changed feature fusion network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, doi: 10.1109/TGRS.2023.3241097.



Xiande Wu received the B.S. degree from Lanzhou University of Technology, Lanzhou, China, in 2018. He is currently pursuing the doctoral degree with the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education, School of Artificial Intelligence, Xidian University, Xi'an, China.

His research interests include deep learning and remote sensing image processing.



Paolo Gamba (Fellow, IEEE) received the Laurea degree (cum laude) and the Ph.D. degree in electronic engineering from the University of Pavia, Italy, in 1989 and 1993, respectively.

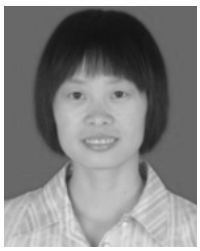
He is a Professor at the Telecommunications and Remote Sensing Laboratory, University of Pavia, Pavia, Italy. He published more than 190 articles in international peer-reviewed journals. He has been invited to give keynote lectures and tutorials on several occasions about urban remote sensing, data fusion, earth observation (EO) data for physical exposure, and risk management.

Dr. Gamba is Fellow of IAPR, AAIA, and the Academia Europaea. He served as Editor-in-Chief of the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, from 2009 to 2013, and as Chair of the Data Fusion Committee of the IEEE Geoscience and Remote Sensing Society (GRSS), from October 2005 to May 2009. He has been elected in the GRSS AdCom from 2014 to 2022, served as GRSS President from 2019 to 2020, and is currently the Editor-in-Chief of the *IEEE Geoscience and Remote Sensing Magazine*. He also served as Technical Co-Chair of the 2010, 2015, and 2020 IGARSS conferences, in Honolulu (Hawaii), Milan (Italy), and online, respectively.



Jie Feng (Senior Member, IEEE) received the B.S. degree from Chang'an University, Xi'an, China, in 2008, and the Ph.D. degree from Xidian University, Xi'an, in 2014.

She is currently a Professor with the Laboratory of Intelligent Perception and Image Understanding, Xidian University. Her research interests include remote sensing image processing, deep learning, and machine learning.



Ronghua Shang (Senior Member, IEEE) received the B.S. degree in information and computation science and the Ph.D. degree in pattern recognition and intelligent systems from Xidian University, Xi'an, China, in 2003 and 2008, respectively.

She is currently a Professor with Xidian University. Her research interests include machine learning, pattern recognition evolutionary computation, image processing, and data mining.



Xiangrong Zhang (Senior Member, IEEE) received the B.S. and M.S. degrees in computer application technology from the School of Computer Science, Xidian University, Xi'an, China, in 1999 and 2003, respectively, and the Ph.D. degree in pattern recognition and intelligent system from the School of Electronic Engineering, Xidian University, in 2006.

She is currently a Professor with the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education, Xidian University. From January 2015 to March 2016, she was a Visiting Scientist with the Computer Science and Artificial Intelligence Laboratory, Massachusetts Institute of Technology. Her research interests include pattern recognition, machine learning, and remote sensing image analysis and understanding.



Licheng Jiao (Fellow, IEEE) received the B.S. degree from Shanghai Jiaotong University, Shanghai, China, in 1982, and the M.S. and Ph.D. degrees from Xi'an Jiaotong University, Xi'an, China, in 1984 and 1990, respectively.

He has authored or coauthored more than 150 scientific articles. His research interests include image processing, natural computation, machine learning, and intelligent information processing. He is in charge of about 40 important scientific research projects and published more than 20 monographs and 100 papers in international journals and conferences.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:49:48 UTC from IEEE Xplore. Restrictions apply.